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Systems and methods for low-light image enhancement

Abstract

A method for enhancing low brightness regions in endoscopic video includes receiving a sequence of video frames captured during an endoscopic procedure. A brightness map is generated for each frame of the sequence of video frames. The brightness map is generated based on smoothing regional brightness in the respective frame based on regional brightness gradients. The brightness map for the respective frame includes scaling weights corresponding to pixels of the respective frame. An enhanced sequence of video frames is generated in real-time by adjusting pixel values of each frame of the sequence of video frames by scaling the pixel values of the respective frame by the scaling weights of the brightness map for the respective frame to enhance brightness of low brightness regions of the respective frame. The enhanced sequence of video frames is displayed in real-time.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/132,915, filed Dec. 31, 2020, the entire contents of which are hereby incorporated by reference herein.

FIELD

(1) The present disclosure is related to medical imaging, and in particular, to image processing for medical imaging.

BACKGROUND

(2) Minimally invasive surgery generally involves the use of a high-definition camera coupled to an endoscope inserted through a small incision into a patient to provide a surgeon with a clear and precise view within the body with minimal tissue damage. The endoscope emits light from its distal end to illuminate the surgical cavity and receives light reflected or emitted by tissue within the surgical cavity through a lens or window located at the distal end of the endoscope. Effectively illuminating the surgical cavity for imaging can be challenging. The confined nature of the surgical cavity and the unidirectional illumination provided by the endoscope often leads to inconsistent illumination of the surgical field. For example, regions closer to the distal end of the endoscope may be brightly illuminated while regions further from the distal end may be poorly illuminated. Poor illumination can lead to poor imaging, with detail of poorly illuminated regions being difficult to discern in the resulting imaging. Efforts to increase image brightness for poorly illuminated regions may cause problems for other regions. For example, increasing the level of illumination and/or increasing the camera gain may lead to oversaturation of regions that are more brightly illuminated.

SUMMARY

(3) According to an aspect, systems and methods enhance low brightness regions in endoscopic

video by generating on a frame-by-frame basis a brightness map that smooths regional brightness based on brightness gradients in an image and enhancing the respective frame by adjusting pixel values based on the brightness map. Pixel values associated with lower brightness regions are increased to a greater degree than pixel values associated with higher brightness regions such that lower brightness regions are enhanced without saturating higher brightness regions. The brightness map smooths regional brightness in regions of low brightness gradients while preserving brightness gradients in regions of high brightness gradients, which preserves detail in the frame. Optionally, the brightness map is generated in low scale space to reduce the processing time for providing frame-by-frame enhancement in real time.

(4) According to an aspects, a method for enhancing low brightness regions in endoscopic video includes receiving a sequence of video frames captured during an endoscopic procedure; generating a brightness map for each frame of the sequence of video frames, wherein the brightness map is generated based on smoothing regional brightness in the respective frame based on regional brightness gradients; generating in real-time an enhanced sequence of video frames by adjusting pixel values of each frame of the sequence of video frames based on the brightness map for the respective frame to enhance brightness of low brightness regions of the respective frame; and displaying the enhanced sequence of video frames in real-time. It is noted that the method concerns the processing of endoscopic images. There is no functional link between the method and effects produced by the endoscope on the body. The endoscope can be pre-inserted into the body. The method can exclude the step of inserting the endoscope into the body. The method is not a method of treatment of the body.

(5) Optionally, the enhanced sequence of video frames is generated in real-time by a graphics processing unit (GPU).

(6) Optionally, the method includes down-sampling the respective frame and generating the respective brightness map based on the down-sampling of the respective frame. The method may include up-sampling the respective brightness map and adjusting the pixel values of the respective frame based on the up-sampling of the respective brightness map. The respective brightness map may be up-sampled using bilinear interpolation.

(7) Optionally, generating the brightness map for each frame of the sequence of video frames includes minimizing a cost function, wherein the cost function comprises a sparse matrix and minimizing the cost function comprises constructing a compressed sparse matrix structure. The compressed sparse matrix structure may be constructed in compressed sparse row format. The compressed sparse matrix structure may include a row index vector, a column index vector, and a non-zero values vector, the row index vector and column index vector may be the same for each frame, and constructing the sparse matrix structure may include updating the non-zero values vector for the respective frame. An iterative conjugate gradient method may be used to solve the minimization of the cost function.

(8) Optionally, the brightness map for the respective frame comprises a matrix of weights for pixels of the respective frame, wherein each weight is computed based on a brightness gradient of a corresponding region of pixels of the respective frame. Adjusting pixel values of each frame may include scaling the pixel values by the weights of the matrix of weights. Each frame may include different color channels and adjusting pixel values of each frame may include scaling each color channel of a given pixel by the same weight.

(9) According to an aspect, a system includes one or more processors, memory, and one or more programs stored in the memory for execution by the one or more processors for: receiving a sequence of video frames captured during an endoscopic procedure; generating a brightness map for each frame of the sequence of video frames, wherein the brightness map is generated based on smoothing of regional brightness in the respective frame based on regional brightness gradients; generating in real-time an enhanced sequence of video frames by adjusting pixel values of each frame of the sequence of video frames based on the brightness map for the respective frame to

enhance brightness of low brightness regions of the respective frame; and displaying the enhanced sequence of video frames in real-time.

(10) Optionally, the enhanced sequence of video frames is generated in real-time by a graphics processing unit (GPU).

(11) Optionally, the one or more programs include instructions for down-sampling the respective frame and generating the respective brightness map based on the down-sampling of the respective frame. The one or more programs may also include instructions for up-sampling the respective brightness map and adjusting the pixel values of the respective frame based on the up-sampling of the respective brightness map. The respective brightness map may be up-sampled using bilinear interpolation.

(12) Optionally, generating the brightness map for each frame of the sequence of video frames includes minimizing a cost function, wherein the cost function comprises a sparse matrix and minimizing the cost function comprises constructing a compressed sparse matrix structure. The compressed sparse matrix structure may be constructed in compressed sparse row format. The compressed sparse matrix structure may include a row index vector, a column index vector, and a non-zero values vector, the row index vector and column index vector may be the same for each frame, and constructing the sparse matrix structure may include updating the non-zero values vector for the respective frame.

(13) Optionally, the minimization of the cost function is solved via an iterative conjugate gradient method.

(14) Optionally, the brightness map for the respective frame comprises a matrix of weights for pixels of the respective frame, wherein each weight is computed based on a brightness gradient of a corresponding region of pixels of the respective frame. Adjusting pixel values of each frame may include scaling the pixel values by the weights of the matrix of weights. Each frame may include different color channels and adjusting pixel values of each frame may include scaling each color channel of a given pixel by the same weight.

(15) According to an aspect, a non-transitory computer readable medium includes instructions for execution by one or more processors for performing any one of the methods above. According to an aspect, a computer program product is provided comprising instructions which, when executed by one or more processors of a computer system, cause the computer system to perform any of the methods above.

(16) It will be appreciated that any of the variations, aspects, features and options described in view of the systems apply equally to the methods and vice versa. It will also be clear that any one or more of the above variations, aspects, features and options can be combined.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the U.S. Patent and Trademark Office upon request and payment of the necessary fee. The invention will now be described, by way of example only, with reference to the accompanying drawings, in which:

(2) FIG. 1 is a block diagram of an exemplary endoscopic imaging system;

(3) FIG. 2 illustrates an exemplary method for low-light image enhancement for endoscopic video imaging;

(4) FIGS. 3A and 3B illustrate an example of a brightness map generated from a video frame;

(5) FIG. 4 illustrates an example of the creation of an enhanced frame from an original frame and a brightness map;

(6) FIG. 5 illustrates an exemplary method for low-light image enhancement;

(7) FIG. 6 illustrates an example of the results of various steps of the method of FIG. 5; and

(8) FIG. 7 illustrates an example of a computing system 700.

DETAILED DESCRIPTION

(9) Reference will now be made in detail to implementations and examples of various aspects and variations of systems and methods described herein. Although several exemplary variations of the systems and methods are described herein, other variations of the systems and methods may include aspects of the systems and methods described herein combined in any suitable manner having combinations of all or some of the aspects described.

(10) According to various aspects, systems and methods can provide brightness enhancement of images (single images or video frames) via a process that adjusts pixel values according to regional brightness. In some examples, a brightness map is generated for the image in which regions of low brightness gradients in the image are represented by regions of relatively uniform values, while the structure areas of strong gradients (e.g., edges) are preserved in the brightness map. The brightness map is used to scale the pixel values of the image, resulting in an enhanced image. In some examples, video frames are enhanced and techniques for reducing memory usage and processing time are used to enable real-time video frame (frame-by-frame) enhancement. In some examples, the image enhancement is performed on a GPU and parallel processing techniques are used to generate the brightness map. In some examples, the brightness map is generated by minimizing a cost function.

(11) In some examples, one or more techniques are used to minimize memory usage and/or reduce processing time so that the brightness map and subsequent image enhancement can be performed in real-time. The brightness map can be generated based on an initial brightness map estimate generated from a respective frame and the initial brightness map estimate is down-sampled so that the cost function minimization problem involves a smaller data set, which can enable real-time video frame enhancement. The data set for the cost minimization problem can be stored in a compressed sparse matrix format, which reduces memory space and/or processing time. In some examples, the cost minimization problem is solved via an iterative conjugate gradient method utilizing a parallel processing platform, such as a graphics processing unit, to enable real-time solutions. The techniques for reducing memory storage and/or processing time may be critical for enabling real-time enhancement of video frames, in particular, for high definition video frames (e.g., 2K, 4K, or higher resolutions).

(12) In the following description, it is to be understood that the singular forms “a,” “an,” and “the” used in the following description are intended to include the plural forms as well, unless the context clearly indicates otherwise. It is also to be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It is further to be understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used herein, specify the presence of stated features, integers, steps, operations, elements, components, and/or units but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, units, and/or groups thereof.

(13) Certain aspects of the present disclosure include process steps and instructions described herein in the form of an algorithm. It should be noted that the process steps and instructions of the present disclosure could be embodied in software, firmware, or hardware and, when embodied in software, could be downloaded to reside on and be operated from different platforms used by a variety of operating systems. Unless specifically stated otherwise as apparent from the following discussion, it is appreciated that, throughout the description, discussions utilizing terms such as “processing,” “computing,” “calculating,” “determining,” “displaying,” “generating” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system memories or registers or other such information storage, transmission, or display devices.

(14) The present disclosure in some examples also relates to a device for performing the operations

herein. This device may be specially constructed for the required purposes, or it may comprise a general-purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, computer readable storage medium, such as, but not limited to, any type of disk, including floppy disks, USB flash drives, external hard drives, optical disks, CD-ROMs, magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, application specific integrated circuits (ASICs), or any type of media suitable for storing electronic instructions, and each coupled to a computer system bus. Furthermore, the computers referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability. Suitable processors include central processing units (CPUs), graphical processing units (GPUs), field programmable gate arrays (FPGAs), and ASICs.

(15) The methods, devices, and systems described herein are not inherently related to any particular computer or other apparatus. Various general-purpose systems may also be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the required method steps. The required structure for a variety of these systems will appear from the description below. In addition, the present invention is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the present invention as described herein.

(16) FIG. 1 illustrates an exemplary system **100** for endoscopic imaging. System **100** includes an endoscope **102** for insertion into a surgical cavity **104** for imaging tissue **106** within the surgical cavity **104** during a medical procedure and an illuminator **120** that provides illumination light to the endoscope **102**, which emits the light from its distal end **114** for illuminating the tissue **106**.

(17) The endoscope **102** may extend from an endoscopic camera head **108** that includes one or more imaging sensors **110**. As is well known in the art, light reflected and/or emitted (such as fluorescence light emitted by fluorescing targets that are excited by fluorescence excitation illumination light) from the tissue **106** is received by the distal end **114** of the endoscope **102**. The light is propagated by the endoscope **102**, such as via one or more optical components (for example, one or more lenses, prisms, light pipes, or other optical components), to the camera head **108**, where it is directed onto the one or more imaging sensors **110**. In some examples, one or more filters (not shown) may be included in the endoscope **102** and/or camera head **108** for filtering a portion of the light received from the tissue **106** (such as fluorescence excitation light).

(18) The one or more imaging sensors **110** generate pixel data that can be transmitted to a camera control unit **112** that is communicatively connected to the camera head **108**. The camera control unit **112** generates one or more images from the pixel data. As used herein “images” encompasses single images and video frames. In some examples, the images can be transmitted to an image processing unit **116** for further image processing, storage, display, and/or routing to an external device (not shown). The images can be transmitted to one or more displays **118**, from the camera control unit **112** and/or the image processing unit **116**, for visualization by medical personnel, such as by a surgeon for visualizing the surgical field **104** during a surgical procedure on a patient.

(19) The illuminator **120** generates illumination light and provides the illumination light to the endoscope **102** via light guide **136**, which may comprise, for example, one or more fiber optic cables, with the light guide **136** coupled to the endoscope **102** at the light post **126** in the proximal region of the endoscope. The illumination light is emitted from the distal end **114** of the endoscope **102** and illuminates the tissue **106**. As used herein, “illumination light” refers to light that may be used to illuminate tissue for the purposes of imaging the tissue. In some examples, at least a portion of the illumination light may be light having a waveband in the visible spectrum that is reflected by the tissue **106** and captured by the one or more imaging sensors **110** for generating visible light imaging data. According to various examples and/or operating modes, at least a portion of the

illumination light can be fluorescence excitation light for exciting one or more fluorescing targets in the tissue, which can include one or more fluorescence agents and/or one or more auto-fluorescing targets. Light emitted by the one or more fluorescence targets may be captured by the one or more imaging sensors for generating fluorescence imaging data. The illumination light can include any desirable wavebands or combination of wavebands.

(20) The illuminator **120** includes one or more light sources **122** that each generate one or more wavebands of light and a controller **124** for controlling the light sources **122**. The light sources **122** generate illumination light for illuminating the tissue **106** via the endoscope **102**. The controller **124** can be configured to activate and deactivate the individual light sources **122** and/or adjust a power level of the light sources **122** to adjust the level of intensity (i.e., the luminance) of light generated by the individual light sources **122**. In some examples, the controller **124** may control one or more light sources **122** based on one or more control signals received from camera control unit **112** and/or image processing unit **116**. In some examples, control signals received from the camera control unit **112** and/or image processing unit **116** can instruct the controller **124** how to control individual light sources **122**, such as by including instructions to activate or deactivate individual light sources **122** and/or to set one or more individual light sources **122** at a specified power/intensity level. For example, the camera control unit **112** and/or image processing unit **116** may determine that a specific color and/or intensity adjustment is needed based on analysis of pixel data generated by the endoscopic camera head **108** and the controller **124** may receive instructions specifying the adjustments.

(21) FIG. 2 illustrates an exemplary method **200** for low-light image enhancement for endoscopic video imaging. Method **200** can be performed by an endoscopic imaging system, such as system **100** of FIG. 1. For example, method **200** can be performed by an image processing unit, such as image processing unit **116** of FIG. 1. In some examples, method **200** or at least a portion of method **200** is performed by a GPU for real-time enhancement and display of video frames.

(22) At step **202**, a sequence of video frames is received. The sequence of video frames may be received from an imager or imaging system or may be received from a memory in which a previously generated sequence of video frames is stored. The sequence of video frames may be received frame-by-frame in a continuous fashion as the frames are generated (i.e., in real-time). In some examples, the sequence of video frames was captured during an endoscopic procedure, which may be an exploratory procedure, a diagnostic procedure, a study, a surgical procedure, a non-surgical procedure, an invasive procedure, or a non-invasive procedure.

(23) At step **204**, a brightness map is generated for each frame of the sequence of video frames received at step **202**. The brightness map is generated based on a smoothing of regional brightness in the frame based on regional brightness gradients in the frame. The brightness of a region having relatively low gradients is smoothed while relatively large brightness gradients are preserved. The brightness map is a matrix of scaling weights corresponding to the pixels of the respective frame. Generally, each weight is computed based on a brightness gradient of a corresponding region of pixels of the respective frame. The weights corresponding to areas of low gradient are substantially similar, whereas weights corresponding to regions of strong gradient preserve a degree of the gradient structure.

(24) FIGS. 3A and 3B illustrate a brightness map generated from a video frame according to various examples of step **204**. FIG. 3A is a video frame **300-A** that is fairly dim, such as due to poor illumination of the field of view. FIG. 3B is a brightness map **300-B** generated from the video frame of FIG. 3A. Comparing region **302-A** in the video frame with region **302-B** in the brightness map, it can be seen that the brightness in the brightness map is smoothed relative to the image frame with little to no texture from region **302-A** being represented in **302-B**. Since region **302-A** in the video frame **300-A** has relatively low brightness gradient, the resulting region **302-B** in the brightness map **300-B** has been smoothed

(25) However, while low brightness gradients are smoothed, high brightness gradients are still

represented in the brightness map. For example, the small dark region **304-A** in the video frame **300-A** just beneath region **302-A** is represented by region **304-B** in the brightness map **300-B** and the transition between the small dark region **304-A** and the surrounding regions is still evident in the region **304-B** and its surrounding regions in the brightness map **300-B**. As another example, a relatively stark bright to dark transition in region **306-A** in the video frame **300-A** is represented in region **306-B** in the brightness map **300-B**. Thus, the brightness map **300-B** preserves the stronger brightness-based “structural” aspects of video frame **300-A**, while smoothing the more minor “textural” aspects of video frame **300-A**.

(26) Returning to FIG. 2, at step **206**, an enhanced sequence of video frames is generated by adjusting pixel values of each frame based on the brightness map for the respective frame. A given pixel value is adjusted based on the corresponding scaling weight of the brightness map. The adjustment enhances brightness of relatively low brightness regions of the respective frame while not saturating regions of relatively high brightness regions. Due to the smoothing of regional brightness in the brightness map for low gradient regions, pixels within the low gradient regions are adjusted by similar scaling weights. But for pixels within the high gradient regions, the scaling weights will be different. Thus, the enhancement can be thought of as a gradient-based regional brightness enhancement. Since pixel values in a region of the video frame are adjusted together, finer (“textural”) details and local contrast are preserved.

(27) Adjustment of pixel values in step **206** is such that pixels in a region where the corresponding brightness map values are uniform will be adjusted by the same scaling weights. In some examples, the adjustment of pixel values in the video frames is inversely proportional to the brightness map such that pixel values corresponding to lower-brightness regions of the brightness map are increased more than pixel values corresponding to higher-brightness regions of the brightness map. In some examples, the same brightness map is used to adjust each color channel of the image.

(28) FIG. 4 illustrates an example of the creation of an enhanced frame **400** from the original frame **300-A** and the brightness map **300-B** from FIG. 3A and FIG. 3B. The brightness of the enhanced frame **400** is greater than that of the original frame **300-A**, with more detail being evident due to the increased brightness. Comparing similar regions of the enhanced frame **400** with the brightness map **300-B**, it is evident that the textural detail is preserved in the enhanced frame for regions where the brightness map has smoothed the brightness gradients. Moreover, it is evident that the brightness of darker regions of the original frame **300-A** has been increased to a greater degree than the brightness of brighter regions.

(29) At step **208**, the enhanced sequence of video frames is displayed in real-time. In other words, the enhancement of the frames is performed as the frames are received. The enhanced frames can be displayed on a display screen during the imaging session, such as intra-operatively in the operating room to assist a surgeon in identifying tissue. In some examples, method **200** may be repeated for each frame in a sequence of video frames. For example, a series of video frames may be generated in real-time during an endoscopic procedure to assist a clinician in visualizing the tissue in the imaging field, and method **200** may be performed on each video frame as the frames are generated. Enhanced images may be generated and displayed as an enhanced video on a display screen to assist the clinician in visualizing the field of view.

(30) Method **200** can include illuminating the tissue with illumination light, capturing the video frames, and transferring the captured video frames from the camera to a system for low-light enhancement of the frames, such as image processing unit **116** of FIG. 1. In some examples, video frames are analyzed to determine whether low-light image enhancement is needed such that method **200** is performed on an ad-hoc basis based on a determination that enhancement is needed. The determination can be made, for example, by image processing unit **116**. In some examples, a user can command image enhancement, such as based on viewing frames that the user considers too dark. In some examples, method **200** is performed regardless of whether enhancement is subjectively needed, for example, because method **200** does not result in saturation and, therefore,

poses little risk for image degradation.

(31) One or more video frames generated by the imager can comprise an intensity value for each pixel of the imager. In some examples, the one or more video frames are based on frames generated by the imager. For example, one or more processing steps may be performed between the generation of the frames by the imager and the performance of step 202 of method 200. Examples of such processing steps include scaling, trimming, denoising, and normalizing. Accordingly, in some examples, the number of intensity values in the frame received at step 202 may be different than the number of pixels of the imager that generated the frame from which the received frame was derived. In some examples, one or more image processing steps may be performed after the frame is received at step 202. For example, generating the enhanced frame may include one or more of a scaling step, a trimming step, a denoising step, and a normalizing step.

(32) FIG. 5 illustrates an exemplary method 500 for low-light image enhancement, according to various aspects of method 200 of FIG. 2. As discussed further below, method 500 implements data conserving techniques to enable low-light image enhancement on high definition video in real-time.

(33) Method 500 generates a brightness map by minimizing the following cost function:

$$(34) \min_T (\|\hat{T} - T\|_F^2 + \sum_{d \in \{h, v\}} W_d(x) \frac{(\nabla_d T(x))^2}{\|\nabla_d \hat{T}(x)\|_2 + \epsilon}) \quad (1)$$

(35) T is the brightness map whose values are refined to minimize expression (1) and $\{\text{circumflex over (T)}\}$ is an initial brightness map estimation. The first term in expression (1) seeks to minimize the difference between the brightness map T and the initial brightness map estimate $\{\text{circumflex over (T)}\}$. The second term seeks to smooth areas of low brightness gradient, while preserving structure related to strong brightness gradients (e.g., edges). $\nabla_{\text{sub}.d}\{\text{circumflex over (T)}\}(x)$ represents the gradient at pixel x in the initial brightness map estimate $\{\text{circumflex over (T)}\}$. $\nabla_{\text{sub}.d}T(x)$ represents the gradient at pixel x in the brightness map T. ϵ is a threshold value that is selected to distinguish strong gradient areas from low gradient areas (e.g., edges from non-edges). An exemplary value for ϵ is 0.01. For areas with high gradients (e.g., strong edges), the gradient is much larger than ϵ , so

$$(36) \frac{(\nabla_d T(x))^2}{\|\nabla_d \hat{T}(x)\|_2 + \epsilon}$$

in equation (1) is approximately close to a constant (e.g. close to 1). In this case, $T \approx \{\text{circumflex over (T)}\}$ and high gradients are preserved. For the areas with minor (textural) details, the gradients of both T and $\{\text{circumflex over (T)}\}$ are less than ϵ , and to minimize the cost function (1), the gradients of T could be minimized since the gradients of $\{\text{circumflex over (T)}\}$ are fixed. In this case, the local textural areas are smoothed and will have substantially consistent brightness level. Thus, expression (1) can provide a brightness map in which low brightness gradient areas are smoothed and strong gradients transitions are preserved.

(37) In expression (1), $W_{\text{sub}.d}(x)$ is a weight. Different weight strategies can be used. In some examples, $W_{\text{sub}.d}(x)$ is a value of 1. In other examples, $W_{\text{sub}.d}(x)$ is a function of $\{\text{circumflex over (T)}\}$, such as follows:

$$(38) \frac{1}{\|\nabla_d \hat{T}(x)\|_2 + \epsilon}.$$

Expression (1) also includes weighting parameter α , which is a predefined value. In some examples, α is in the range of 0.01 to 10.

(39) Minimizing the cost function of expression (1) is equivalent to solving the following linear system ($Ax=b$):

$$(40) (I + \sum_{d \in \{u, v\}} D_d^T \text{Diag}(\tilde{w}_d) D_d) t = \hat{t} \quad (2)$$

(41) Returning to FIG. 5, at step 502, a video frame is received. At step 504, an initial brightness map estimate $\{\text{circumflex over (T)}\}$ is generated. In some examples, the initial brightness map is

the relative luminance of the video frame, calculated from the RGB values of the video frame, according to well-known methods. In some examples, the initial brightness map is the maximum value of the RGB values for a given pixel. In some examples, the initial brightness map value for a given pixel is based on the maximum value of a region of pixels centered on the given pixel.

(42) At step **506**, the initial brightness map estimate $\{\text{circumflex over (T)}\}$ is down sampled to reduce processing time in minimizing expression (1) since minimizing expression (1) for full-scale high resolution images (e.g., 2K and 4K resolution) may not be feasible in real-time on commercial image processing systems (e.g., even using processors tailored to image processing such as GPUs). In some examples, the brightness map initial brightness map estimate $\{\text{circumflex over (T)}\}$ is down sampled by a factor in the range from 0.1 to 0.5. In some examples, step **506** is not performed.

(43) The matrix in equation (2) is a sparse matrix with dimension $N \times N$, where $N = R \times C$, and R is the row number and C is the column number of the image frame. For high resolution images, the size of this matrix presents a memory storage and/or processing challenge for solving equation (2) in a timely fashion, such as required for real-time processing. The inventors realized that there are only $5N$ non-zero elements in the sparse matrix. Hence, according to various examples, method **500** includes optional step **508** in which the sparse matrix structure of equation (2) can be constructed in a compressed sparse matrix structure. In some examples, the compressed sparse matrix structure is a compressed sparse row (CSR) format to store the non-zero elements only in the sparse matrix. In some examples, the CSR format includes: Row index vector of $N+1$ elements; Column index vector of $5N$ elements; and Non-zero values vector of $5N$ elements. Accordingly, only $11N+1$ space may be required to store the $N \times N$ sparse matrix of equation (2), which represents a significant savings in stored and processed data so that equation (2) can be solved in real-time for high resolution video frames, according to various examples.

(44) For example, given a small image with resolution 128×128 , the sparse matrix is 16384×16384 , which even for the small image may be still too big to be stored in GPU memory. However, for this example with the CSR sparse matrix structure, only $11 \times 16384 + 1$ space is needed, which saves 99.93% of the memory required for the original matrix. Even compared with the coordinate list (COO) format ((row column, value) tuples with $15N$ spaces), it also saves 26.67% of the memory space.

(45) In some examples, other sparse matrix structures may be used to reduce the storage space and processing time for solving equation (2), such as COO and compressed sparse column (CSC)

(46) In some examples, once the sparse matrix structure of equation (2) is initialized (e.g., for the first frame in the series of video frames) in the CSR format, the row index and column index vectors are constant for all subsequent image frames. The only additional step is to update the non-zero values item in the structure based on the new frame. This could be used for estimating the brightness map directly, which can speed up the real-time processing.

(47) At step **510**, an iterative conjugate gradient method is used to solve the sparse linear system of equation (2) (with or without the compressed structure format of step **506**). Since the matrix in equation (2) is a symmetric and positive-definite sparse matrix, the solution t is unique. Therefore, a conjugate gradient method can be used to iteratively estimate the solution, such as on a parallel computing platform (e.g., a GPU). In some examples, leveraging the CSR storage format of step **506**, the cuSparse and cuBLAS APIs in CUDA can be used to speed up the calculation in parallel for solving equation (2) in real-time, such as on a GPU. Step **510** provides the brightness map T .

(48) At step **512**, the brightness map is up-sampled if the initial brightness map estimate was down-sampled in step **506**. In some examples, the brightness map is up-sampled to the original resolution via bilinear interpolation. In some variations, the brightness map may be up-sampled to the original resolution via bicubic interpolation or another interpolation method may be used for the up-sampling.

(49) At optional step **514**, a gamma correction is applied to the brightness map, with $T \leftarrow T \cdot \text{sup} \cdot \gamma$.

Any suitable value for γ may be used, such as any value in the range of 0.1 to 1.0.

(50) At step **516**, the original video frame is enhanced via the brightness map (with or without the gamma correction). Each pixel of the original video frame is scaled according to the corresponding weight of the brightness map. In some examples, due to the brightness map smoothing small gradients, pixel values in a low gradient region may be scaled by equivalent or substantially equivalent weights even though the pixel values may be different. As such, areas are brightened as a region, which preserves detail. In some examples, the original video frame may be enhanced by dividing each pixel by the corresponding weight from the brightness map. In some examples, the same brightness map weight is used to adjust each color channel for each pixel of a color frame. In some examples, one or more additional post-processing steps may be performed, such as one or more denoising and/or recomposing steps which may be performed to produce the final enhanced frame.

(51) Method **500** can be performed on each frame of a sequence of video frames to generate an enhanced sequence of video frames. In some examples, method **500** can be performed in real-time to generate enhanced video frames in real-time (i.e., at the same rate that the video frames are captured). In some examples, method **500** is performed via a parallel processing platform. In some examples, method **500** is performed on a GPU. In some examples, method **500** is performed on high resolution video frames (e.g., 2K, 4K, or higher resolution video frames).

(52) FIG. **6** illustrates an example of the results of various steps of method **500**. The input frame **600** is used to generate an initial brightness map estimate **602**, which as discussed above, can be a maximum value of the RGB channels for a respective pixel or the relative luminance Y for a respective pixel. A brightness map **604** is generated from the initial brightness map estimate **602**, such as via solving equation (2) and using one or more of steps **506-514** of method **500**. An enhanced frame **606** is generated from the brightness map **604** by scaling each pixel by the respective weight of the brightness map.

(53) FIG. **7** illustrates an example of a computing system **700** that can be used for one or more of components of system **100** of FIG. **1**, such as one or more of camera head **108**, camera control unit **112**, image processing unit **116**, and illuminator **120**. System **700** can be a computer connected to a network, such as one or more networks of hospital, including a local area network within a room of a medical facility and a network linking different portions of the medical facility. System **700** can be a client or a server. As shown in FIG. **7**, system **700** can be any suitable type of processor-based system, such as a personal computer, workstation, server, handheld computing device (portable electronic device) such as a phone or tablet, or dedicated device. The system **700** can include, for example, one or more of input device **720**, output device **730**, one or more processors **710**, storage **740**, and communication device **760**. Input device **720** and output device **730** can generally correspond to those described above and can either be connectable or integrated with the computer.

(54) Input device **720** can be any suitable device that provides input, such as a touch screen, keyboard or keypad, mouse, gesture recognition component of a virtual/augmented reality system, or voice-recognition device. Output device **730** can be or include any suitable device that provides output, such as a display, touch screen, haptics device, virtual/augmented reality display, or speaker.

(55) Storage **740** can be any suitable device that provides storage, such as an electrical, magnetic, or optical memory including a RAM, cache, hard drive, removable storage disk, or other non-transitory computer readable medium. Communication device **760** can include any suitable device capable of transmitting and receiving signals over a network, such as a network interface chip or device. The components of the computing system **700** can be connected in any suitable manner, such as via a physical bus or wirelessly.

(56) Processor(s) **710** can be any suitable processor or combination of processors, including any of, or any combination of, a central processing unit (CPU), graphics processing unit (GPU), field programmable gate array (FPGA), and application-specific integrated circuit (ASIC). Software

750, which can be stored in storage **740** and executed by one or more processors **710**, can include, for example, the programming that embodies the functionality or portions of the functionality of the present disclosure (e.g., as embodied in the devices as described above). For example, software **750** can include one or more programs for execution by one or more processor(s) **710** for performing one or more of the steps of method **200** and/or method **500**.

(57) Software **750** can also be stored and/or transported within any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a computer-readable storage medium can be any medium, such as storage **740**, that can contain or store programming for use by or in connection with an instruction execution system, apparatus, or device.

(58) Software **750** can also be propagated within any transport medium for use by or in connection with an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a transport medium can be any medium that can communicate, propagate or transport programming for use by or in connection with an instruction execution system, apparatus, or device. The transport computer readable medium can include, but is not limited to, an electronic, magnetic, optical, electromagnetic, or infrared wired or wireless propagation medium.

(59) System **700** may be connected to a network, which can be any suitable type of interconnected communication system. The network can implement any suitable communications protocol and can be secured by any suitable security protocol. The network can comprise network links of any suitable arrangement that can implement the transmission and reception of network signals, such as wireless network connections, T1 or T3 lines, cable networks, DSL, or telephone lines.

(60) System **700** can implement any operating system suitable for operating on the network. Software **750** can be written in any suitable programming language, such as C, C++, Java, or Python. In various examples, application software embodying the functionality of the present disclosure can be deployed in different configurations, such as in a client/server arrangement or through a Web browser as a Web-based application or Web service, for example.

(61) The foregoing description, for the purpose of explanation, has been described with reference to specific examples. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The examples were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various examples with various modifications as are suited to the particular use contemplated.

(62) Although the disclosure and examples have been fully described with reference to the accompanying figures, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims. Finally, the entire disclosure of the patents and publications referred to in this application are hereby incorporated herein by reference.

Claims

1. A method for enhancing low brightness regions in endoscopic video comprising: receiving a sequence of video frames captured during an endoscopic procedure; generating a brightness map for each frame of the sequence of video frames, wherein the brightness map is generated based on smoothing regional brightness in the respective frame based on regional brightness gradients, and

wherein the brightness map for the respective frame comprises scaling weights corresponding to pixels of the respective frame; generating in real-time an enhanced sequence of video frames as the sequence of video frames is received during the endoscopic procedure by adjusting pixel values of each frame of the sequence of video frames by scaling the pixel values of the respective frame by the scaling weights of the brightness map for the respective frame to enhance brightness of low brightness regions of the respective frame; and displaying the enhanced sequence of video frames to a surgeon in real-time during the endoscopic procedure.

2. The method of claim 1, wherein the enhanced sequence of video frames is generated in real-time by a graphics processing unit (GPU).

3. The method of claim 1, comprising down-sampling the respective frame and generating the respective brightness map based on the down-sampling of the respective frame.

4. The method of claim 3, comprising up-sampling the respective brightness map and adjusting the pixel values of the respective frame based on the up-sampling of the respective brightness map.

5. The method of claim 4, wherein the respective brightness map is up-sampled using bilinear interpolation.

6. The method of claim 1, wherein generating the brightness map for each frame of the sequence of video frames comprises minimizing a cost function, wherein the cost function comprises a sparse matrix and minimizing the cost function comprises constructing a compressed sparse matrix structure.

7. The method of claim 6, wherein the compressed sparse matrix structure is constructed in compressed sparse row format.

8. The method of claim 6, wherein the compressed sparse matrix structure includes a row index vector, a column index vector, and a non-zero values vector, the row index vector and column index vector are the same for each frame, and constructing the sparse matrix structure comprises updating the non-zero values vector for the respective frame.

9. The method of claim 6, comprising using an iterative conjugate gradient method to solve the minimization of the cost function.

10. The method of claim 1, wherein each of the scaling weights is computed based on a brightness gradient of a corresponding region of pixels of the respective frame.

11. The method of claim 10, wherein adjusting the pixel values of each of the frames comprises scaling the pixel values by the weights of the matrix of weights.

12. The method of claim 10, wherein each of the frames comprises different color channels and adjusting pixel values of each of the frames comprises scaling each of the color channels of a given pixel by the same weight.

13. The method of claim 1, comprising, prior to generating a brightness map for each frame of the sequence of video frames: generating an initial brightness map estimate for each frame of the sequence of video frames; and down-sampling the initial brightness map estimate for each frame of the sequence of video frames.

14. The method of claim 13, wherein the brightness map for each frame of the sequence of video frames is generated based on the respective down-sampled initial brightness map estimate.

15. A system comprising one or more processors, memory, and one or more programs stored in the memory for execution by the one or more processors for: receiving a sequence of video frames captured during an endoscopic procedure; generating a brightness map for each frame of the sequence of video frames, wherein the brightness map is generated based on smoothing regional brightness in the respective frame based on regional brightness gradients, and wherein the brightness map for the respective frame comprises scaling weights corresponding to pixels of the respective frame; generating in real-time an enhanced sequence of video frames as the sequence of video frames is received during the endoscopic procedure by adjusting pixel values of each frame of the sequence of video frames by scaling the pixel values of the respective frame by the scaling weights of the brightness map for the respective frame to enhance brightness of low brightness

regions of the respective frame; and displaying the enhanced sequence of video frames to a surgeon in real-time during the endoscopic procedure.

16. The system of claim 15, wherein the enhanced sequence of video frames is generated in real-time by a graphics processing unit (GPU).

17. The system of claim 15, wherein the one or more programs include instructions for down-sampling the respective frame and generating the respective brightness map based on the down-sampling of the respective frame.

18. The system of claim 17, wherein the one or more programs include instructions for up-sampling the respective brightness map and adjusting the pixel values of the respective frame based on the up-sampling of the respective brightness map.

19. The system of claim 18, wherein the respective brightness map is up-sampled using bilinear interpolation.

20. The system of claim 15, wherein generating the brightness map for each frame of the sequence of video frames comprises minimizing a cost function, wherein the cost function comprises a sparse matrix and minimizing the cost function comprises constructing a compressed sparse matrix structure.

21. The system of claim 20, wherein the compressed sparse matrix structure is constructed in compressed sparse row format.

22. The system of claim 20, wherein the compressed sparse matrix structure includes a row index vector, a column index vector, and a non-zero values vector, the row index vector and column index vector are the same for each frame, and constructing the sparse matrix structure comprises updating the non-zero values vector for the respective frame.

23. The system of claim 20, wherein the minimization of the cost function is solved via an iterative conjugate gradient method.

24. The system of claim 15, wherein each of the scaling weights is computed based on a brightness gradient of a corresponding region of pixels of the respective frame.

25. The system of claim 24, wherein adjusting the pixel values of each of the frames comprises scaling the pixel values by the weights of the matrix of weights.

26. The system of claim 24, wherein each of the frames comprises different color channels and adjusting pixel values of each of the frames comprises scaling each of the color channels of a given pixel by the same weight.

27. The system of claim 15, wherein the one or more programs include instructions for, prior to generating a brightness map for each frame of the sequence of video frames: generating an initial brightness map estimate for each frame of the sequence of video frames; and down-sampling the initial brightness map estimate for each frame of the sequence of video frames.

28. The system of claim 27, wherein the brightness map for each frame of the sequence of video frames is generated based on the respective down-sampled initial brightness map estimate.

29. A non-transitory computer readable medium comprising instructions for execution by one or more processors of a system to cause the system to: receive a sequence of video frames captured during an endoscopic procedure; generate a brightness map for each frame of the sequence of video frames, wherein the brightness map is generated based on smoothing regional brightness in the respective frame based on regional brightness gradients, and wherein the brightness map for the respective frame comprises scaling weights corresponding to pixels of the respective frame; generate in real-time an enhanced sequence of video frames as the sequence of video frames is received during the endoscopic procedure by adjusting pixel values of each frame of the sequence of video frames by scaling the pixel values of the respective frame by the scaling weights of the brightness map for the respective frame to enhance brightness of low brightness regions of the respective frame; and display the enhanced sequence of video frames to a surgeon in real-time during the endoscopic procedure.
